

Ben M. Andrew

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Education

University of Manchester, PhD in Computer Science Sep 2024 – Present

- Project titled *Developing Reliable Software for Autonomous Robots*.
- Supervised by Dr. Marie Farrell, Dr. Louise Dennis, and Prof. Michael Fisher.

University of Cambridge, BA (Hons) in Computer Science Oct 2019 – Jun 2022

- Grade II.1 (70%).
- Dissertation titled *Delay-Tolerant Link-State Routing*. Implemented and empirically evaluated a failure-tolerant routing protocol in C to run on real Unix-based routers, comparing with traditional protocols. Supervised by Dr. Jackson Woodruff.

Research Interests

My interests lie in the intersection of autonomous robotics and formal methods. Specifically, I am passionate about verifying mission-critical systems under uncertainty and failure.

Publications

- Andrew, B. M., Dennis, L. A., Fisher, M., Farrell, M. *Counterexample-Guided Interval Weakening*. Rigorous State-Based Methods (ABZ) 2026. To Appear.
- Andrew, B. M. *Weakening Goals in Logical Specifications*. Rigorous State-Based Methods (ABZ) 2025. 10.1007/978-3-031-94533-5_22

Professional Activities

- Part of the local organising committee of Integrated Formal Methods (iFM) 2024.
- Member of the Autonomy and Verification Network.

Teaching Experience

University of Cambridge, Supervisor (TA equiv.)

- Part II CST: Hoare Logic and Model Checking 2022 – 2026
- Part IB CST: Logic and Proof 2022 – 2026
- Part IB CST: Semantics of Programming Languages 2023
- Part IA CST: Introduction to Graphics 2022, 2025

University of Manchester, Graduate Teaching Assistant

- COMP23412: Software Engineering 2025
- COMP60332: Automated Reasoning and Verification 2024
- COMP21111: Logic and Modelling 2024
- COMP11120: Mathematical Techniques for Computer Science 2024

Industry Experience

Software Engineer, Tarides – Paris, France Sep 2022 – Jun 2024

- Developed open-source continuous integration (CI) software for the OCaml language's ecosystem, involving systems programming to distribute dependency solving, compilation, and testing to many operating systems and architectures.
- Responsible for the CI of the Opam Repository, the central package universe for OCaml with tens of thousands of packages.

Embedded Systems Intern, Tarides – Paris, France

Jul – Sep 2021

- Research & development project to cross-compile the OCaml bytecode runtime to an ARM-based microcontroller using a real-time operating system as a base layer.

Computer Graphics Research Intern, University of Cambridge

Jun – Sep 2020

- Developed and implemented a visual model of ageing vision in virtual reality with Unity, as part of the Graphics and Displays Research Group.

Volunteering

Run Leader, Runwild – University of Manchester

Dec 2024 – Present

- Lead bi-weekly group runs for a student running group.

English Teacher, Connections English Institute – Ibagué, Colombia

Jul – Aug 2024

- Taught adult beginners in morning and night classes.